

Uncertainty-Aware Multimodal Deep Learning Framework for Stress Detection Using Audio-Visual Signals

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Abstract—Abstract-Stress detection has become an important research due to the increasing commonness of mental health issues which are stress related and on overall productivity and well being. The traditional stress assessment method are relied upon on physiological signals or self-prepared questionnaires, which might not always provide reliable results or can't help in continuous monitoring. Recent advancements in deep learning methods have enabled the use of multimodal data, such as audio, video and also physiological data, which helps in automatic recognition of stress.

In this paper, we propose an Uncertainty-Aware Hybrid Deep Learning framework (UA-HybridNet) for detection of stress via multimodal audio-video data. The proposed model architecture includes convolutional neural networks (CNN) for spatial feature extraction and recurrent neural networks for temporal modeling, enabling effective combination of complementary information from the speech and facial expressions. Unlike conventional method, the proposed model has uncertainty estimation and probability calibration techniques to improve robustness and reliability of prediction output.

Upon experimenting the evaluation demonstrated that the proposed framework achieves a competitive perform of 81.3% accuracy and an AUC of 0.878, while also providing uncertainty-aware predictions and calibrated confidence estimates. These capabilities help with the reliability of the model when dealing with real-world stress monitoring scenarios, where it is critical to make trustworthy-decisions.

The results of the uncertainty-aware multimodal deep learning highlight potential for building reliable and scalable stress detection systems for healthcare industries and applications which require human-computer interaction.

Keywords: Stress Detection, Multimodal Learning, Deep Learning, Uncertainty Estimation, Audio-Visual Analysis

I. INTRODUCTION

Stress is a rapidly growing concern in our modern society, especially in work and clinical environments which are very stressful, there may be people of anxiety disorders, depression, cardiovascular issues, and impaired cognitive functioning under the impact of psychological pressures. Continuous stress not only affects individual well-being, but also impacts their professional productivity and decision-making ability. Early detection and accurate categorization of state of the stress are

therefore essential for preventive healthcare, workplace safety, and mental health intervention strategies [1]–[3].

Regular stress assessment methods, such as self-prepared questionnaires, interviews, and psychological systems for scoring, often suffer from subjectivity, delayed reports and limited real-time application [4], [5]. These methods are dependent heavily on subjective perception and may not capture minute physiological and behavioral changes involved with stress reactions. So as a result, it is clear that there is need for automated, objective, and real-time systems for stress detection, which can be continuously monitored.

Recent improvements in deep learning and affective computing have enabled the development of intelligent stress recognition systems that analyze multimodal signals, including facial expressions and speech patterns. Audio-visual datasets such as RAVDESS facilitate structured modeling of emotion-related stress states through synchronized speech and facial recordings [5]. Deep neural networks, particularly convolutional and spatiotemporal architectures, have demonstrated strong capabilities for extracting discriminative representations from facial dynamics and acoustic features [6]–[8]. Such analytical methods improve the accuracy of stress detection and decrease biases of subjective estimations.

To improve multimodal prediction performance, this study uses a hybrid deep learning system consisting of 3D convolutional neural networks with spatial-temporal facial modeling, EfficientNet-B0 with high-level spatial feature distribution, an Action Unit (AU) gating system, and an audio convolutional branch to enhance multimodal prediction. In contrast to conventional deterministic classifiers, the given system also relies on uncertainty-aware mechanisms to enhance reliability and deployment safety.

The proposed framework widens regular multimodal stress classification by combining with Monte Carlo Dropout for predictive uncertainty estimation, temperature scaling for confidence calibration, test-time augmentation for robustness enhancement, and selective prediction for abstention on ambiguous samples. Monte Carlo Dropout approximates Bayesian inference through stochastic forward passes, allowing computation of predictive mean and variance. Temperature scaling

corrects overconfident probability estimates without affecting classification accuracy. Selective prediction mechanisms allow the system to reject high-uncertainty predictions, thereby the reliability is improved in safety-critical environments.

The experimental analysis is a meticulous comparative analysis of the baseline CNN-LSTM, the transformer-based cross-attention, and the suggested hybrid architecture. Findings indicate that the process of recalling stress detection is relatively robust. Although deterministic models achieve high accuracy, incorporating uncertainty estimation significantly improves reliability and decision confidence and calibration, thereby boosting model trustworthiness, decreasing overconfidence, and increasing selective performance when uncertainty is considered. This research advances the development of reliable AI-powered stress monitoring systems that can produce calibrated probabilistic outputs and make decisions that account for uncertainty. These systems are especially useful for telemedicine platforms, preventive mental health screening, and occupational stress monitoring. Future studies could investigate real-time deployment with edge-based multimodal sensing systems, conformal prediction frameworks, and out-of-distribution detection. Although multimodal stress detection and uncertainty estimation have been studied separately in the past, their integration for calibrated, selective stress prediction has received little attention. By integrating risk-aware selective prediction, probabilistic calibration, and hybrid deep feature learning into a single framework, UA-HybridNet fills this gap. This connection improves deployment dependability in safety-critical stress-monitoring scenarios in addition to classification capabilities. The main contributions of this work are:

- An uncertainty-aware multimodal stress detection framework combining 3D-CNN, EfficientNet-B0, and AU gating.
- Integration of Monte Carlo Dropout and temperature scaling for joint uncertainty estimation and probability calibration.
- Comprehensive evaluation using calibration metrics, uncertainty AUROC, and risk-coverage analysis.
- Selective prediction strategy to improve reliability in safety-critical stress monitoring applications.

II. RELATED WORKS

Stress detection based on Artificial intelligence and machine learning has attracted serious interest, mainly in monitoring of professional work and in healthcare. Gedam and Paul [1] studied wearable sensor-based stress-detection techniques that use physiological signals such as skin conductance and heart rate. Despite its effectiveness, these methods require constant device use and are also prone to inter-individual variability and noise. Mentis et al. [2] highlighted the use of deep learning models for complex behavioral and physiological analysis of data, which highlights the expanding relevance of AI in applications on mental health. Similarly, multimodal physiological datasets for continuous monitoring of stress were introduced by Hosseini et al. [3] and Iqbal et al. [4]; however, both of them rely mainly on specialized sensing equipment.

An additional medium which can be used for analyzing stress and emotions is audio-visual data. For recognizing multimodal emotions, the RAVDESS dataset [5] provides synchronized voice and facial data. Deep learning architectures such as EfficientNet [6] and spatiotemporal convolutional networks [7] have displayed very strong capacity to extract selective visual and temporal features. Sequential data modeling was further improved by attention-based models, mainly by the Transformer architecture [8], through relational learning. Deep learning models are successful, but they give overconfident, settled predictions very often. To improve uncertainty estimations and registration, Monte Carlo Dropout [9] and temperature scaling [10] were added. Kendall and Gal [11] highlighted the importance of capturing both data noise and model uncertainty by differencing between aleatoric and epistemic uncertainty in the vision systems. Deep ensemble techniques were presented by Lakshminarayanan et al. [12], which improves predictive strength by using various independently trained networks. To improve consistency, Geifman and El-Yaniv [13] proposed selective categorization algorithms which allows models from avoiding making low-confidence predictions. Similar methods for identifying misclassified and out-of-distribution samples were proposed by Hendrycks and Gimpel [14], which is necessary for the implementation of stress detection systems practically. Recently there was a research by Nag et al. [15], which shows that advanced optimization strategies can improve stress prediction in occupational situations.

Furthermore, reliable AI systems require clear decision boundaries, well-calibrated confidence estimates, and accurate predictions. The ability of a model to identify unknown or ambiguous inputs is crucial in safety-critical applications such as healthcare stress monitoring. These uncertainty-aware techniques enable the development of dependable multimodal stress detection frameworks suitable for real-world implementation.

A. Research Gap

Despite these significant advancements, there still remains several restrictions. Wearable physiological sensors are a major component of many stress detection systems, which restricts scalability and practical implementation. Furthermore, most of the deep learning techniques which are in use now, give out deterministic results without any reliable uncertainty estimation. To improve the dependability and practical applicability, we require a multimodal stress detection system that combines audio-visual data with uncertainty-aware methods.

III. STRESS PREDICTION MODEL OVERVIEW

A. Purpose

For the proposed approach our objective was to provide an accurate multimodal framework for automatic stress detection using facial expression and audio data. Traditional unimodal systems for stress detection are often seen to perform poorly because they rely solely on visual or auditory data and

the contextual cues are lacking. To overcome this limitation, the proposed approach combines speech-based acoustic representations with face video data using a hybrid deep learning architecture. The system uses EfficientNet-B0 to extract high-level spatial representations from facial frames and a 3D Convolutional Neural Network (3D-CNN) to record spatial-temporal facial dynamics. Because of their compound scaling strategy [6], EfficientNet architectures have achieved strong performance in visual recognition challenges. The framework aims to enhance stress recognition performance and provide more reliable classification of stress levels by combining visual and auditory modalities.

B. System Workflow

Figure 1 illustrates the proposed multimodal stress detection framework. Data collection, preprocessing, feature extraction, multimodal fusion, and classification are all a structured part of the system's pipeline. The RAVDESS dataset, which contains the emotional speech data as audio-video recordings, it is the primary data source. The dataset includes organized facial video frames and relative voice signals [5]. To make sure the input data is high-quality, the preprocessing techniques such as frame extraction, face detection, audio normalization, and temporal segmentation are used. To process the facial frames for visual analysis, a hybrid deep learning architecture is used which combines 3D-CNN and EfficientNet-B0 to capture both temporal motion dynamics and spatial facial representations. Correspondingly, the audio characteristics which are associated with Stress are captured from speech data by extracting acoustic components. A multimodal feature fusion approach is utilized to combine the extracted visual and auditory data. For the final categorization, fully connected layers are applied to the fused model. Two categories: *Stress* and *Non-Stress* are predicted by the output layer. By leveraging complementary information across multiple modalities, multimodal learning techniques have demonstrated enhanced robustness and classification performance [2].

C. Dataset

The publicly available RAVDESS (Ryerson Audio-Visual Database of Emotional Speech and Song) dataset, which is frequently utilized in emotion identification research, is employed in this study. The dataset contains audio-visual recordings of emotional speech performed by 24 professional actors (12 male and 12 female) [5]. Eight different emotional states, neutral, calm, happy, sad, furious, afraid, disgusted, and surprised, are expressed by each actor. High-quality video footage, synced with speech signals, enhances the recordings. For the detection of stress, emotions like anger, fear, and sadness are categorized as stress-related, whereas for non-stress states, calm and neutral expressions are classified in this category. RAVDESS contains controlled, high-quality, synchronized audio and video recordings that are appropriate for testing multimodal learning architectures, even though they involve performed emotional expressions rather than spontaneous real-world stress signals. The structured nature

of the dataset enables systematic evaluation of calibration and uncertainty mechanisms. To further test the generalization capabilities, future work will expand UA-HybridNet to real-world stress datasets and in-the-wild recordings.

D. Data Preprocessing

It is also important to preprocess effectively to enhance model performance and to achieve reliable stress detection. The preprocessing pipeline comprises the facial frame extraction, face detection, image resizing, and normalization. Video clips are decomposed into a sequence of frames, and facial areas are recognized and isolated. All the pictures are downsized to a similar resolution to allow the use of deep learning models. In the case of a sound channel, speech sounds are normalized and converted to Mel-spectrogram representations. These time-frequency representations capture significant acoustic features of emotional stress and speech dynamics.

E. Data Segmentation

The breaking up of video scenes into smaller pieces in motion-related data is captured using Temporal segmentation. The 3D-CNN model can capture temporal shifts in facial expressions because in each video, the sample is divided into multiple frame sequences of fixed-length. Similarly, the audio signals are also divided into brief time intervals to show spectrogram descriptions that capture the dynamics of speech, which is related to stress.

F. Feature Extraction Using Deep Learning

Deep learning architectures that automatically learns selective features on visual and acoustic modalities.

• 3D Convolutional Neural Network (3D-CNN)

The 3D-CNN model extracts spatio-temporal features from facial video sequences. In contrast to other 2D CNNs, 3D CNNs combine spatial and temporal dimensions of convolution, which allows the model to learn patterns of facial expressions [7].

$$F_{out}(x, y, t) = \sum_i \sum_j \sum_k W(i, j, k) \cdot X(x+i, y+j, t+k) \quad (1)$$

where X represents the input feature volume and W denotes the learnable 3D convolution kernel.

• EfficientNet-B0

From the facial frame the high-level spatial features are extracted using the EfficientNet-B0. EfficientNet employs compound scaling to balance network depth, width, and input resolution, enabling efficient feature learning with fewer parameters [6].

$$Depth = \alpha^\phi, \quad Width = \beta^\phi, \quad Resolution = \gamma^\phi \quad (2)$$

subject to

$$\alpha \cdot \beta^2 \cdot \gamma^2 \approx 2 \quad (3)$$

where ϕ controls the scaling coefficient of the network.

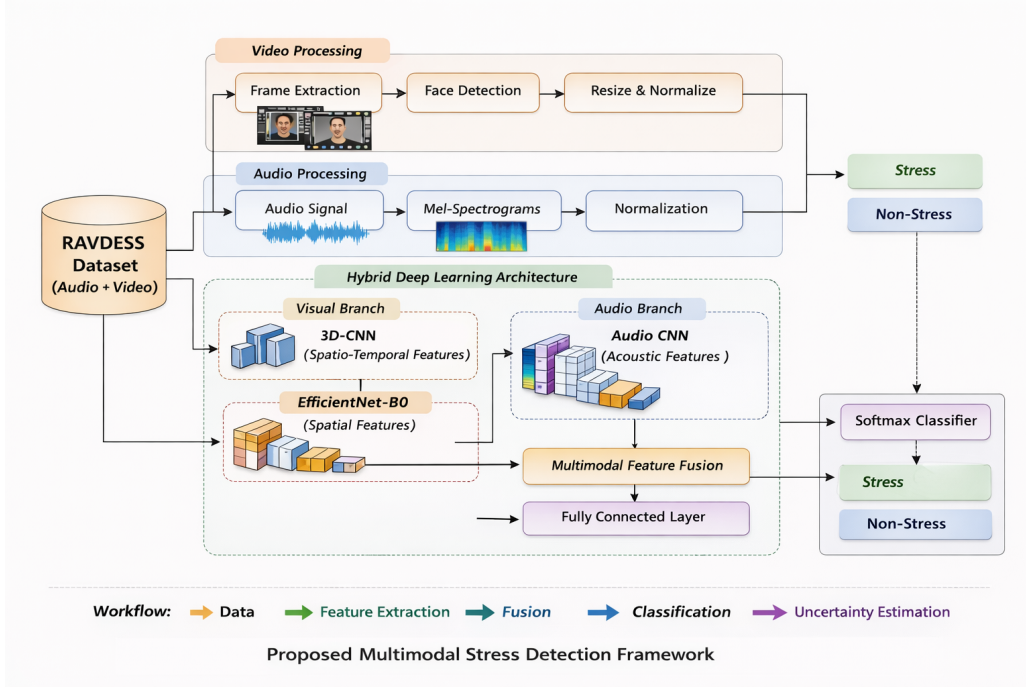


Fig. 1: Proposed Multimodal Stress Detection Framework

G. Multimodal Feature Fusion

In order to use the complementary information provided by the speech and facial modalities, we have used Feature-level fusion. The visual and audio feature vectors are obtained and merged together to have a unified representation.

$$F_{fusion} = [F_{visual} \oplus F_{audio}] \quad (4)$$

where $\oplus$ represents feature concatenation.

To utilise its multimodal representation, the model is capable of obtaining dynamics of facial expressions and speech characteristics that are concerned with stress. Multimodal fusion techniques have found wide application in affective computing to enhance the robustness and reliability of prediction [2].

H. Model Training and Stress Classification

To classify the fused feature vector, we have applied fully connected layers. The final prediction layer predicts the probability distribution of stress classes using the SoftMax activation function.

$$P(y = k|x) = \frac{e^{z_k}}{\sum_{j=1}^K e^{z_j}} \quad (5)$$

Two classification, *Stress* and *Non-Stress*, corresponds to the final output as a task of binary classification. During training, to optimize the model's parameters we have used backpropagation. Standard metrics, such as accuracy, precision, recall, F1-score, and Area Under the Curve (AUC), are used to evaluate performance and are also frequently used to assess the stress detection systems in machine learning research.

IV. RESULTS AND DISCUSSION

This section provides a comprehensive analysis of **UA-HybridNet**, which combines uncertainty estimation and probability calibration with the Hybrid (3D-CNN + EfficientNet-B0 + AU) architecture.

1) *Overall Classification Performance*: A relative performance analysis of all explored architectures, including the baseline CNN-LSTM model, the cross-attention dual-stream model, the deterministic Hybrid architecture, and the suggested framework, is shown in Table I. The deterministic Hybrid model includes probabilistic calibration and uncertainty estimates, which significantly improves the dependability and interpretability of predictions, even though the achieved raw accuracy is slightly higher. So as a result, the proposed methodology places more importance on reliable predictions than just focusing on point-estimate accuracy.

TABLE I: Performance Comparison Across All Architectures

Model	Acc.	Prec.	Recall	F1	AUC	Reliability
Baseline (CNN-2D + LSTM)	0.885	0.860	0.920	0.889	0.971	No
Cross-Attention (Dual-Stream)	0.790	0.790	0.790	0.790	0.890	No
Hybrid (3D-CNN + EfficientNet-B0 + Aux.)	0.850	0.802	0.930	0.861	0.940	No
Hybrid + Uncertainty Modeling	0.813	0.783	0.867	0.823	0.878	Yes

Even though the deterministic Hybrid model gives higher raw accuracy, UA-HybridNet prefers computed, uncertainty-aware predictions over point-estimate optimization. In health-care stress monitoring areas of safety-critical applications, overconfidence and wrong estimates can have serious consequences. So as a result, reliability, compute quality and selective control of risk are more important than minor

improvements in raw accuracy. The results show that UA-HybridNet significantly improve confidence arrangement and risk-aware decision making while maintaining competitive level classification performance.

To illustrate this trade-off further, Table II gives a direct comparison between deterministic Hybrid model and the uncertainty-aware model. The results show that, despite getting considerably higher precision, the deterministic model gives calibrated confidence estimates, which are necessary in safety-critical applications for reliable deployment.

TABLE II: Classification Performance: Hybrid vs UA-HybridNet

Metric	Hybrid	UA-HybridNet
Accuracy	0.850	0.813
Recall	0.930	0.867
F1	0.861	0.823
AUC	0.940	0.878

2) *Per-Class Performance*: For a balanced test set of 150 samples (75 stress and 75 non-stress) the per-class classification report is shown in Table III. While maintaining balanced macro and weighted averages across precision, recall, and F1-score, for the non-stress class, the results show slightly better detection performance.

TABLE III: TABLE III: Per-Class Classification Report for UA-HybridNet

Class	Precision	Recall	F1 Score	Support
Stress	0.8500	0.7600	0.8000	75
Non-Stress	0.7800	0.8700	0.8200	75
Macro Avg	0.8200	0.8100	0.8100	150
Weighted Avg	0.8200	0.8100	0.8100	150

A wide perspective of classification results given by the confusion matrix is shown in Fig. 2. Out of the 150 test samples, 28 are incorrectly classified, and 122 are correctly classified.

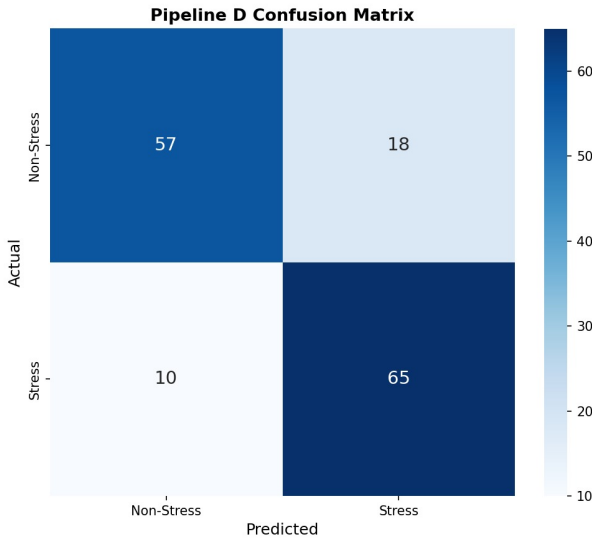


Fig. 2: Confusion Matrix for .

3) *Calibration Performance*: The crucial part of probability calibration is Temperature scaling. Calibration metrics are shown in Table IV, of both before and after scaling with a learning temperature parameter of $T = 3.0855$. By reducing the Expected Calibration Error (ECE) by 60% and Maximum Calibration Error (MCE) by 67%, the quality of calibration is greatly improved by temperature scaling.

TABLE IV: Calibration Metrics Before and After Temperature Scaling

Metric	Before	After	Improvement
ECE	0.1768	0.0710	↓ 59.8%
MCE	0.7651	0.2531	↓ 66.9%
Brier Score	0.1734	0.1450	↓ 16.4%
NLL	0.7172	0.3940	↓ 45.1%

Figure 3 further shows this improvement using reliability diagrams of before and after calibration.

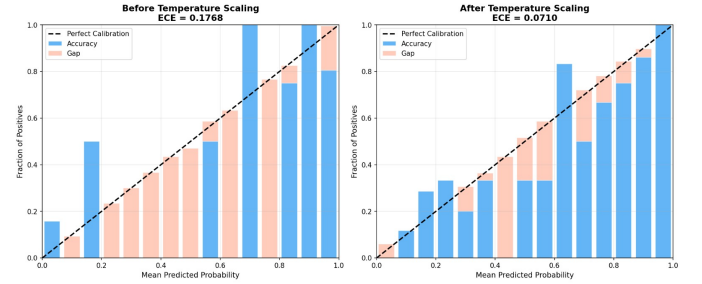


Fig. 3: Reliability diagrams before and after temperature scaling.

4) *Uncertainty Estimation Analysis*: Monte Carlo Dropout with $T = 20$ stochastic forward passes is used to estimate uncertainty. Key uncertainty measures, such as prediction variance, entropy, and the Uncertainty AUROC, are summarized in Table V.

TABLE V: Uncertainty Estimation Statistics

Metric	Value
Mean Predictive Variance	0.001564
Mean Predictive Entropy	0.1092
Uncertainty AUROC	0.6906
Risk-Coverage AUC	0.1015

The misclassified samples generally show more predictive variance than correctly classified samples, which was confirmed by the uncertainty distribution shown in Fig. 4

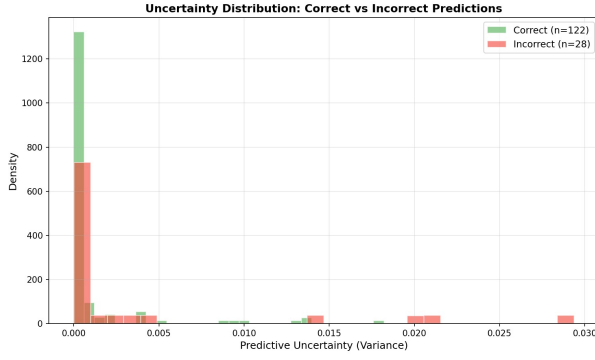


Fig. 4: Predictive uncertainty distribution for correct and incorrect predictions.

5) *Selective Prediction Performance*: Selective prediction experiments assess the effects of rejecting samples that may not be certain. Increasing the rejection rate progressively enhances classification performance, as Table VI demonstrates.

TABLE VI: Selective Prediction Performance at Different Rejection Levels

Rejection Rate	Kept	Rejected	Accuracy	Precision	Recall	F1 Score	AUC
0%	150	0	0.8133	0.7831	0.8667	0.8228	0.8782
5%	143	7	0.8322	0.8077	0.8750	0.8400	0.8834
10%	135	15	0.8296	0.8108	0.8696	0.8392	0.8887
20%	120	30	0.8333	0.8088	0.8871	0.8462	0.8999

The risk-coverage curve in Fig. 5 shows that rejecting high-uncertainty samples significantly reduces prediction error.

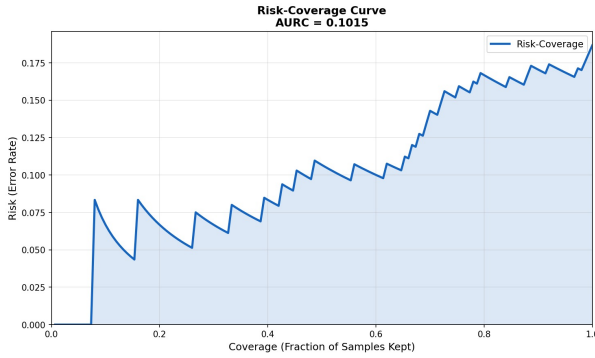


Fig. 5: Risk-Coverage Curve demonstrating uncertainty-aware error reduction.

6) *Clinical Safety Analysis*: Under uncertainty constraints the Table VII evaluates the stress recall. Excluding the uncertain samples improves the recall for stress detection, an important factor in the applications of safety-sensitive monitoring.

TABLE VII: Clinical Safety: Stress Recall under Uncertainty Constraints

Condition	Recall (Stress)	Samples Evaluated
All samples	0.8667	150
Uncertainty < median	0.8750	143
Uncertainty < 75th percentile	0.8696	135
Uncertainty < 80th percentile	0.8871	120

Overall, the results show that UA-HybridNet successfully involves uncertainty estimates and computation into deep multimodal feature learning, allowing reliable detection of stress with robustness and prediction with higher confidence. The CNN-LSTM baseline gives deterministic predictions without confidence or quantifying the uncertainty of prediction, but achieves the highest possible classification of raw accuracy. By combining uncertainty estimation and probability calibration techniques, the suggested UA-HybridNet framework mainly focuses on reliability-aware modeling. Through computed estimates of confidence and selective prediction of risk-awareness, this combination greatly increases the reliability of prediction with a small reduction in the point-estimate accuracy. In application of safety-sensitive stress monitoring, where overconfident, incorrect predictions can lead to misleading decision, hence these characteristics are especially important.

A. Limitations

The proposed framework demonstrates encouraging performance; however, it still has several drawbacks. First, applicability to real-world settings may be impacted since the RAVDESS dataset includes performed emotional expressions rather than spontaneous real-world stress signals. Secondly, compared with large-scale deep learning datasets, the dataset is rather modest in size. Larger multimodal datasets and real-time deployment situations will be the focus of future research.

V. CONCLUSION

To detect the facial stress, this research proposed an uncertainty-aware deep learning system which combines probabilistic calibration and uncertainty estimation with hybrid feature learning. The proposed model involves Action Unit gating with the Monte-Carlo Dropout and temperature scaling, EfficientNet-B0 spatial feature extraction, and 3D-CNN spatiotemporal encoding. A very strong classification performance is shown by the experimental findings on the RAVDESS dataset, with an accuracy of 0.8133 and an AUC of 0.8782. By reducing the negative log-likelihood by 45.1% and the Expected Calibration Error (ECE) from 0.1768 to 0.0710, the calibration is greatly improved by temperature scaling and showed an increased agreement between actual accuracy and expected confidence. Selective prediction further improved the system dependability: accuracy increased to 0.8333 and AUC to 0.8999 at a 20% rejection rate. These results show that uncertainty estimation improves the overall reliability and successfully detects the incorrect predictions. Thus, the proposed model gives a useful and reliable foundation for application requiring human-computer interactions, evaluation of workplace well-being, and stress monitoring in healthcare.

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